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Session Task: Designing and Identifying Study Types in Clinical and Neuroimaging Research

Task 1: Study Type Identification

For each scenario below, identify the most appropriate study design from the following list:

- Case Report
- Case Series
- Cross-sectional (Descriptive or Analytical)
- Case-Control
- Cohort (Prospective or Retrospective)
- RCT (Parallel, Cross-over, Factorial, or Cluster)
- Community Trial
- Field Trial

Scenarios:

1. Scenario 1:**Retrospective case control**
Researchers compare CSF amyloid-beta levels in 100 Alzheimer's patients with 100 healthy controls. Data are extracted from medical records over the past 5 years.
2. Scenario 2:**Case Report**
A 45-year-old patient with rare visual hallucinations and behavioral changes is reported. An MRI reveals a lesion in the temporal lobe.
3. Scenario 3:**Community Trial**
Researchers implement a meditation program in 5 schools and compare student stress levels with 5 other schools that did not receive the program.
4. Scenario 4:**RCT Parallel**
A clinical trial where depressed patients are randomly assigned to receive either Drug A or Drug B, and treatment effects are compared after 8 weeks.
5. Scenario 5:**Prospective Cohort**
Researchers follow 2,000 healthy individuals for 10 years to determine who develops Parkinson's disease and examine its association with caffeine consumption.

Task2: Finding Systematic Reviews in Neuroscience

Based on the concepts from your session on study types and systematic reviews, this task requires you to apply your knowledge to real-world neuroscience research.

Choose a neuroimaging-related health topic of interest (e.g., functional connectivity in depression, brain changes in chronic pain, of ADHD, mindfulness and brain plasticity, or the effects of antiepileptic drugs on brain structure), and find at least three systematic reviews (at least one meta-analysis) on that topic from different databases.

Criterion	Review 1 (Augmented Reality Interface for Complex Anatomy Learning in the Central Nervous System: A Systematic Review)	Review 2 (Dense longitudinal neuroimaging reveals individual brain change trajectories)	Review 3 (The Neural Correlates of Response Inhibition With and Without Conflict in ADHD: An Activation Likelihood Estimation Meta-Analysis)
Type of Review	Systematic Review.	Narrative Review	Meta-Analysis
authors and year	Rahmita Wirza¹ , Shah Nazir² , Habib Ullah Khan³ , Iván García-Magariño⁴ , Rohul	Sophia Vinci-Booher ¹ , Xueying Ren ¹ , Kendrick Kay² , Chen Yu ³ , Franco Pestilli³ , James R. Booth¹ May 2026	Sarah Davidi¹ , Valerio Santangelo^{1,2} 2025 Sep

	Amin [§] 2020 Sep		
Main Research Question	• R.Q.1. What are the applications of augmented reality in complex anatomy learning for the central nervous system? • R.Q.2. What research work is performed since 2009 in the area of gesture interaction-based augmented reality approaches for complex anatomy learning in the central nervous system? • R.Q.3. What are the available techniques/ methods being used for gesture interaction-based augmented reality approaches for complex anatomy learning?	How can dense longitudinal neuroimaging (DLN) help examine rapid trajectories of brain change and provide insight into the mechanisms underlying these changes?	What are the neural correlates of response inhibition in children and adolescents with ADHD, and do they differ between inhibition with and without conflict?
Databases Searched	a. IEEE b. PubMed c. ScienceDirect d. SpringerLink e. Taylor and Francis Online f. Wiley Online Library	Not reported	Pubmed, Scopus, and Web of Science

Methods Section	The guidelines [16] were followed as protocol for conducting the proposed research work. The first phase of the SLR is to describe questions which consist of three research questions. The second phase is the process of search in the mentioned libraries for the keywords defined. The next phase is the selection process in which the relevant papers are included while the irrelevant papers are excluded. In the fourth phase, the quality assessment process is performed, and weights are assigned to questions. Last, data analysis of the obtainable data in included articles is done.	Not explicitly reported	The researchers conducted a systematic literature search in PubMed, Scopus, and Web of Science. They screened the identified studies based on predefined criteria and included 30 studies. They then used an Activation Likelihood Estimation (ALE) meta-analysis to identify brain regions consistently associated with response inhibition in children and adolescents with ADHD.
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Task 3: Formulating a Research Question with PICO

Scenario Description

A research team is investigating the neural correlates of **inhibitory control deficits** in individuals with **Post-Traumatic Stress Disorder (PTSD)**. They recruit two groups of participants: one consisting of adult patients diagnosed with PTSD, and another consisting of healthy adults with no history of trauma or psychiatric disorders. All participants undergo **fMRI scanning** while performing a **Go/No-Go task** (a classic measure of inhibitory control). The researchers want to explore whether brain activation patterns during successful inhibition differ between the PTSD group and the healthy control group, and whether these differences correlate with symptom severity.

1. Write a research question for this scenario. Do individuals with PTSD show different brain activation patterns during successful response inhibition compared with healthy controls?

2. Define the PICO / PECO Elements

Expected Output:

PICO Element	Description
P (Population)	one consisting of adult patients diagnosed with PTSD
I / E (Intervention / Exposure)	All participants undergo fMRI scanning while performing a Go/No-Go task
C (Comparison)	another consisting of healthy adults with no history of trauma or psychiatric disorders
O (Outcome)	The researchers want to explore whether brain activation patterns during successful inhibition differ between the PTSD group and the healthy control group

3. Identify the Main Concepts

Extract the key concepts from your PICO/PECO elements that will be used in your literature search.

4. Create a Concept Table

Use PubMed (MeSH) to find controlled vocabulary terms, and also collect synonyms, alternate spellings, and related terms.

Concepts	MeSH Terms	Synonyms
P=PTSD	"stress disorders, post traumatic"[MeSH Terms] OR "stress disorders, post traumatic"[MeSH Terms] OR "stress disorders, post traumatic"[MeSH Terms]	post-traumatic stress disorder posttraumatic stress disorder PTSD

I/E=Go/No-Go task	((Go/No-Go task) OR (response inhibition task)) OR (go/no-go paradigm)	Go/No-Go task Go-NoGo task go/no-go paradigm response inhibition task
C= Inhibitory Control	(((((motor inhibition[MeSH Terms]) OR (cognitive inhibition[MeSH Terms])) OR (inhibitory function[MeSH Terms])) OR (inhibition control[MeSH Terms])) OR (response inhibition[MeSH Terms])) OR (inhibitory control[MeSH Terms])	inhibitory control response inhibition inhibition control inhibitory function cognitive inhibition motor inhibition
O= Brain Activation / fMRI	(((((fMRI[MeSH Terms]) OR (brain activity[MeSH Terms])) OR (brain activation[MeSH Terms])) OR (functional MRI[MeSH Terms])) OR (functional neuroimaging[MeSH Terms])) OR (functional magnetic resonance imaging[MeSH Terms])	fMRI functional MRI functional magnetic resonance imaging functional neuroimaging brain activation neural activation brain activity neural activity
In total	(((((((((fMRI[MeSH Terms]) OR (brain activity[MeSH Terms]) OR (brain activation[MeSH Terms]) OR (functional MRI[MeSH Terms]) OR (functional neuroimaging[MeSH Terms]) OR (functional magnetic resonance imaging[MeSH Terms]) AND ((((((motor inhibition[MeSH Terms]) OR (cognitive inhibition[MeSH Terms]) OR (inhibitory function[MeSH Terms]) OR (inhibition control[MeSH Terms]) OR (response inhibition[MeSH Terms]) OR (inhibitory control[MeSH Terms])) AND (((Go/No-Go task) OR (response inhibition task)) OR (go/no-go paradigm))) AND (((post-traumatic stress disorder[MeSH Terms]) OR (posttraumatic stress disorder[MeSH Terms]) OR (PTSD[MeSH Terms]))	